2310:

```
farsi@Ubuntu:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable ../build/scratch/ns3.43-first-default
At time +2s client sent 2310 bytes to 10.1.1.2 port 9
At time +2.00578s server received 2310 bytes from 10.1.1.1 port 49153
At time +2.00578s server sent 2310 bytes to 10.1.1.1 port 49153
At time +2.01156s client received 2310 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 2338
  Rx Bytes: 2338
  Mean Delay: 0.0057792 s
  Throughput: 1039.11 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 2338
  Rx Bytes: 2338
  Mean Delay: 0.0057792 s
  Throughput: 1039.11 bps
```

1510:

```
farsi@Ubuntu:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable ../build/scratch/ns3.43-first-default
At time +2s client sent 1510 bytes to 10.1.1.2 port 9
At time +2.0045s server received 1510 bytes from 10.1.1.1 port 49153
At time +2.0045s server sent 1510 bytes to 10.1.1.1 port 49153
At time +2.009s client received 1510 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 1538
  Rx Bytes: 1538
  Mean Delay: 0.0044992 s
  Throughput: 683.556 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 1538
  Rx Bytes: 1538
  Mean Delay: 0.0044992 s
  Throughput: 683.556 bps
```

132:

```
farsi@Ubuntu:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable ../build/scratch/ns3.43-first-default
At time +2s client sent 132 bytes to 10.1.1.2 port 9
At time +2.00226s server received 132 bytes from 10.1.1.1 port 49153
At time +2.00226s server sent 132 bytes to 10.1.1.1 port 49153
At time +2.00452s client received 132 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 160
  Rx Bytes: 160
  Mean Delay: 0.0022592 s
  Throughput: 71.1111 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 160
  Rx Bytes: 160
  Mean Delay: 0.0022592 s
  Throughput: 71.1111 bps
```

151:

```
farsi@Ubuntu:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable ../build/scratch/ns3.43-first-default
At time +2s client sent 151 bytes to 10.1.1.2 port 9
At time +2.00229s server received 151 bytes from 10.1.1.1 port 49153
At time +2.00229s server sent 151 bytes to 10.1.1.1 port 49153
At time +2.00458s client received 151 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 179
  Rx Bytes: 179
  Mean Delay: 0.0022896 s
  Throughput: 79.5556 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 179
  Rx Bytes: 179
  Mean Delay: 0.0022896 s
  Throughput: 79.5556 bps
```

Code:

```
/*
 * SPDX-License-Identifier: GPL-2.0-only
 */

#include "ns3/applications-module.h"
#include "ns3/core-module.h"
#include "ns3/internet-module.h"
#include "ns3/network-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/flow-monitor-module.h"
//include the above line at the top of the code


// Default Network Topology
//
//       10.1.1.0
// n0 -------------- n1
//       point-to-point
//

using namespace ns3;

NS_LOG_COMPONENT_DEFINE("FirstScriptExample");

int
main(int argc, char* argv[])
{
        CommandLine cmd(__FILE__);
        cmd.Parse(argc, argv);
```

```
Time::SetResolution(Time::NS);
LogComponentEnable("UdpEchoClientApplication", LOG_LEVEL_INFO);
LogComponentEnable("UdpEchoServerApplication", LOG_LEVEL_INFO);

NodeContainer nodes;
nodes.Create(2);

PointToPointHelper pointToPoint;
pointToPoint.SetDeviceAttribute("DataRate", StringValue("5Mbps"));
pointToPoint.SetChannelAttribute("Delay", StringValue("2ms"));

NetDeviceContainer devices;
devices = pointToPoint.Install(nodes);

InternetStackHelper stack;
stack.Install(nodes);

Ipv4AddressHelper address;
address.SetBase("10.1.1.0", "255.255.255.0");

Ipv4InterfaceContainer interfaces = address.Assign(devices);

UdpEchoServerHelper echoServer(9);

ApplicationContainer serverApps = echoServer.Install(nodes.Get(1));
serverApps.Start(Seconds(1.0));
serverApps.Stop(Seconds(10.0));

UdpEchoClientHelper echoClient(interfaces.GetAddress(1), 9);
echoClient.SetAttribute("MaxPackets", UintegerValue(1));
echoClient.SetAttribute("Interval", TimeValue(Seconds(1.0)));
echoClient.SetAttribute("PacketSize", UintegerValue(151));

ApplicationContainer clientApps = echoClient.Install(nodes.Get(0));
clientApps.Start(Seconds(2.0));
clientApps.Stop(Seconds(10.0));

// 1. Setup FlowMonitor
FlowMonitorHelper flowHelper;
Ptr<FlowMonitor> monitor = flowHelper.InstallAll();

// 2. Run Simulation
Simulator::Stop(Seconds(20.0));
Simulator::Run();

// 3. Process Statistics
monitor->CheckForLostPackets();
```

```
        Ptr<Ipv4FlowClassifier> classifier =
DynamicCast<Ipv4FlowClassifier>(flowHelper.GetClassifier());
        std::map<FlowId, FlowMonitor::FlowStats> stats = monitor->GetFlowStats();

        for (auto const& [id, stat] : stats) {
        Ipv4FlowClassifier::FiveTuple t = classifier->FindFlow(id);

        std::string proto = (t.protocol == 6) ? "TCP" : (t.protocol == 17) ? "UDP" : "Unknown";

        std::cout << "FlowID: " << id << " (" << proto << " "
        << t.sourceAddress << "/" << t.sourcePort << " --> "
        << t.destinationAddress << "/" << t.destinationPort << ")\n";

        std::cout << "  Tx Bytes: " << stat.txBytes << "\n";
        std::cout << "  Rx Bytes: " << stat.rxBytes << "\n";

        if (stat.rxPackets > 0) {
        std::cout << "  Mean Delay: " << stat.delaySum.GetSeconds() / stat.rxPackets << "
s\n";
        std::cout << "  Throughput: " << (stat.rxBytes * 8.0) / 18.0 << " bps\n";
        }
        }

        Simulator::Destroy();
        return 0;
}
```

Throughput VS Packet size Graph

